

Advanced (Habanero)

- Q1.** What is the purpose of a scatter plot?
- (A) To show the relationship between two variables
 - (B) To compare the means of two groups
 - (C) To display the distribution of a single variable
 - (D) To test a hypothesis about a population
 - (E) To create a line graph
- Q2.** Which type of correlation is shown by the data points in the graph below?
- (A) Positive correlation
 - (B) Negative correlation
 - (C) No correlation
 - (D) Strong positive correlation
 - (E) Strong negative correlation
- Q3.** What is the equation of the line of best fit for the data (2, 3), (4, 5), (6, 7)?
- (A) $y = x + 1$
 - (B) $y = x - 1$
 - (C) $y = 2x + 1$
 - (D) $y = 2x - 1$
 - (E) $y = x - 2$
- Q4.** If a line of best fit has a slope of 2 and passes through the point (1, 3), what is its equation?
- (A) $y = 2x + 1$
 - (B) $y = 2x - 1$
 - (C) $y = 2x + 3$
 - (D) $y = x + 2$
 - (E) $y = x + 3$
- Q5.** What does the slope of the line of best fit represent?
- (A) The rate of change of the dependent variable
 - (B) The rate of change of the independent variable
 - (C) The correlation coefficient
 - (D) The mean of the data points
 - (E) The standard deviation of the data points
- Q6.** If the line of best fit for a set of data has a slope of -2 and a y -intercept of 5, what is its equation?
- (A) $y = -2x + 5$
 - (B) $y = 2x + 5$
 - (C) $y = -2x - 5$
 - (D) $y = x - 2$
 - (E) $y = x + 5$
- Q7.** What is the purpose of using a line of best fit to model a set of data?
- (A) To find the mean of the data points
 - (B) To find the standard deviation of the data points
 - (C) To make predictions about future data points
 - (D) To compare the means of two groups
 - (E) To test a hypothesis about a population
- Q8.** If a scatter plot shows a strong positive correlation, what can be concluded about the relationship between the variables?
- (A) As one variable increases, the other variable decreases
 - (B) As one variable increases, the other variable also tends to increase
 - (C) There is no relationship between the variables
 - (D) The relationship between the variables is random
 - (E) The variables are unrelated

Open Ended Questions (FRQ)

- Q9.** Using the data points (1, 2), (2, 3), (3, 5), create a scatter plot and draw a line of best fit. Then, use the line of best fit to predict the value of y when $x = 4$. Show your work and explain your reasoning.
- Q10.** Explain the differences between a positive correlation, a negative correlation, and no correlation. Provide examples of each and discuss the implications of each type of correlation.

Chef's Tips

- Tip 1: Emphasize the importance of labeling the axes and title of the scatter plot
- Tip 2: Encourage students to use a ruler to draw the line of best fit
- Tip 3: Remind students to show their work and explain their reasoning for the free response questions